JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI MURUGA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

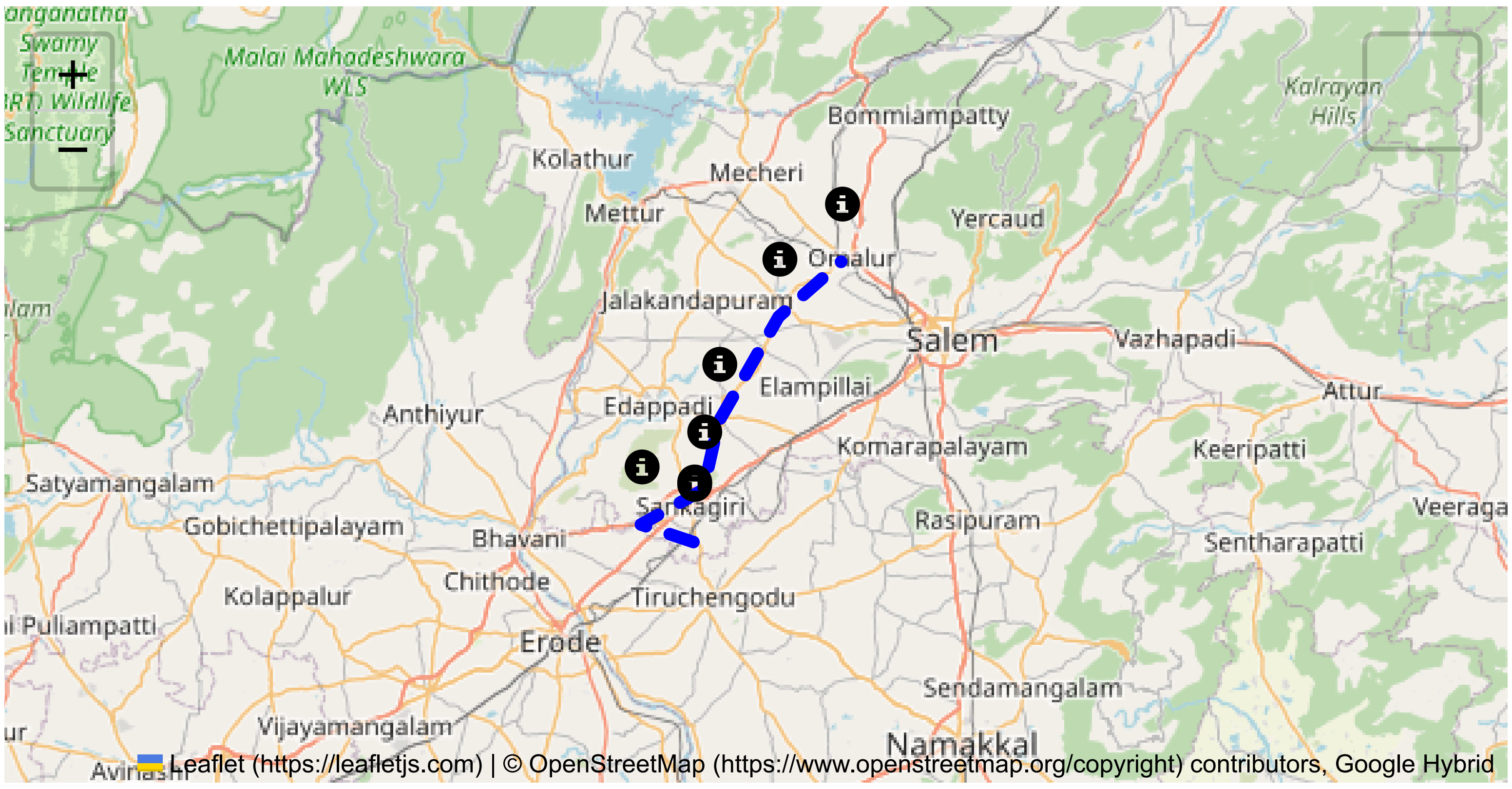
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

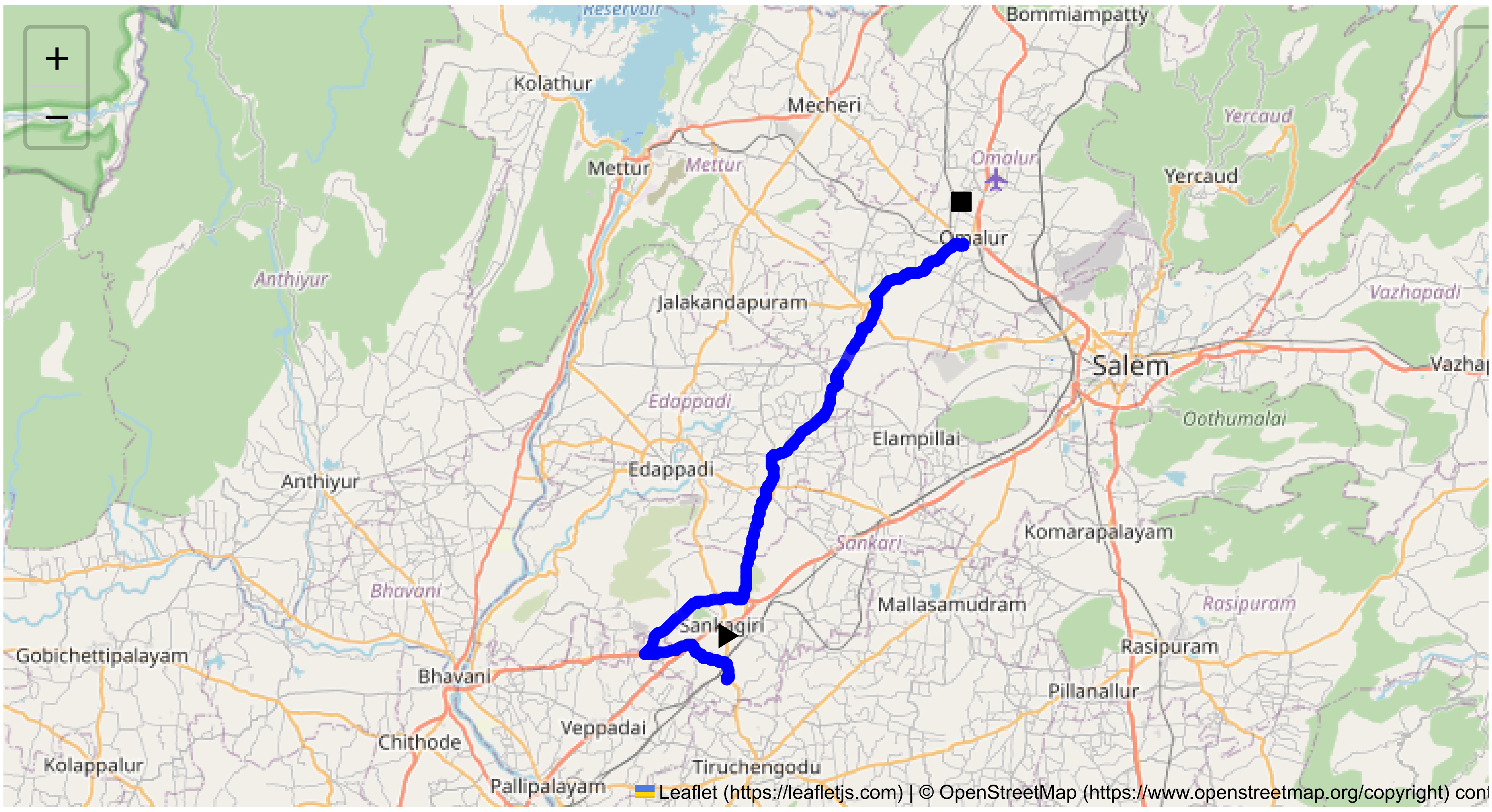
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 53.38 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (11.737009, 78.037086)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route in question travels from Sangagiri to Omalur, spanning approximately 53.38 kilometers. It incorporates segments of local roads and major highways, such as the Salem-Kochi Highway. The journey features a mix of rural and semi-urban stretches with significant agricultural regions through the areas surrounding Sangagiri, Puthur, Padaiveedu, Erumaipatti, and Desavilakku.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. The key weather-related hazards include:

- **Monsoons (June to September)**: Heavy rainfall could lead to water logging and slippery roads, particularly in rural areas.
- **Summer Heat (March to May)**: High temperatures may affect vehicle performance and increase overheating risks.
- **Fog** in the early mornings during winter months (November to February) can reduce visibility.

3. Analysis of Traffic Patterns

- **Peak Hours**: 8:00 AM - 10:00 AM and 5:00 PM - 7:00 PM on weekdays.
- **Congestion-Prone Areas**: The junctions at major highways and market areas, particularly at Sangagiri and Omalur Main Road, may experience higher traffic density and delays.

4. Assessment of Road Quality and Infrastructure

- **Local Roads**: Varying conditions, with some stretches potentially having potholes, especially after monsoon. Narrow roads may limit heavy vehicle passage.
- **Highways (Salem-Kochi, Bhavani Main Road)**: Generally well-maintained, with moderate to good surface quality.

5. Suggestions for Alternative Routes for Emergencies

In case of significant roadblocks or traffic disruptions:

- **Alternative Roadway**: Consider rerouting through NH44, which runs parallel and may provide more reliable passage.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has specific regulations regarding the transport of hazardous materials requiring:

- Prior permissions and notifications.
- Adherence to speed limits specific to load type.
- Prohibition of transport during certain hours in heavily populated areas or sensitive zones.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Accidents involving heavy vehicles** often occur around sharp turns or during overtaking maneuvers on narrow roads.

- **Hazardous Spills**: There have been isolated incidents of chemical spills in more industrial zones; these were typically attributed to inadequate safety measures and poor road conditions.

8. Environmental Considerations and Sensitive Areas

- **Forest and Wildlife**: The route nearby forest patches, requiring caution to protect local flora and fauna.
- **Water Bodies**: Potential contamination risks during accidental spills.

9. Analysis of Communication Coverage

- **Urban Areas**: Generally strong mobile network coverage.
- **Rural Sections**: Potential dead zones may exist at lower altitudes or amidst dense vegetation. It is advisable to check signal strength periodically.

10. Estimated Emergency Response Times

- In populated areas like Sangagiri and Omalur: Approximately 20-30 minutes.
- In rural or semi-urban stretches: Responses may take up to 60 minutes due to accessibility constraints.

11. Overall Summary of Risk Assessment

Overall, the route presents:

- **Moderate Risk** due to variable road conditions, peak traffic congestion, and potential weather impacts.
- **Increased Risk for Hazardous Materials** due to regulatory adherence and environmental sensitivity.
- **Emergency Preparedness**: Ensure communications are functional and emergency contact details are easily accessible.

Drivers are advised to be cautious particularly during peak hours and inclement weather, prepare for poor road conditions, and adhere to safety protocols strictly when transporting hazardous materials. Regular updates on weather and traffic conditions can aid in reducing travel risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43961, 77.87341	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
3	Turn	Medium	11.44898, 77.87410	30 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	Medium	11.45352, 77.85698	30 KM/Hr
6	Turn	Medium	11.46048, 77.85036	30 KM/Hr
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr
8	Turn	Medium	11.46314, 77.84928	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45631, 77.81668	15 KM/Hr
11	Turn	High	11.49409, 77.88004	15 KM/Hr
12	Turn	Medium	11.49419, 77.88015	30 KM/Hr
13	Turn	High	11.49383, 77.88442	15 KM/Hr
14	Turn	Medium	11.49419, 77.88454	30 KM/Hr
15	Turn	Medium	11.49448, 77.88490	30 KM/Hr
16	Turn	High	11.56083, 77.89921	15 KM/Hr
17	Turn	Medium	11.57048, 77.90224	30 KM/Hr
18	Turn	Medium	11.58304, 77.90656	30 KM/Hr
19	Turn	Medium	11.63840, 77.94820	30 KM/Hr
20	Turn	Medium	11.67858, 77.96834	30 KM/Hr
21	Turn	Medium	11.67870, 77.96846	30 KM/Hr
22	Turn	Medium	11.68905, 77.97736	30 KM/Hr
23	Turn	Medium	11.68913, 77.97734	30 KM/Hr
24	Turn	Medium	11.68925, 77.97741	30 KM/Hr
25	Turn	High	11.68927, 77.97747	15 KM/Hr
26	Turn	High	11.70223, 77.97860	15 KM/Hr
27	Turn	Medium	11.71394, 77.99572	30 KM/Hr
28	Turn	Medium	11.73661, 78.03864	30 KM/Hr
29	Turn	Medium	11.73661, 78.03886	30 KM/Hr
30	Turn	High	11.73772, 78.04016	15 KM/Hr
31	Blind Spot	Blind Spot	11.73763, 78.04027	10 KM/Hr
32	Blind Spot	Blind Spot	11.73751, 78.04016	10 KM/Hr
33	Blind Spot	Blind Spot	11.73756, 78.04012	10 KM/Hr
34	Turn	Medium	11.73643, 78.03869	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46048, 77.85036

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

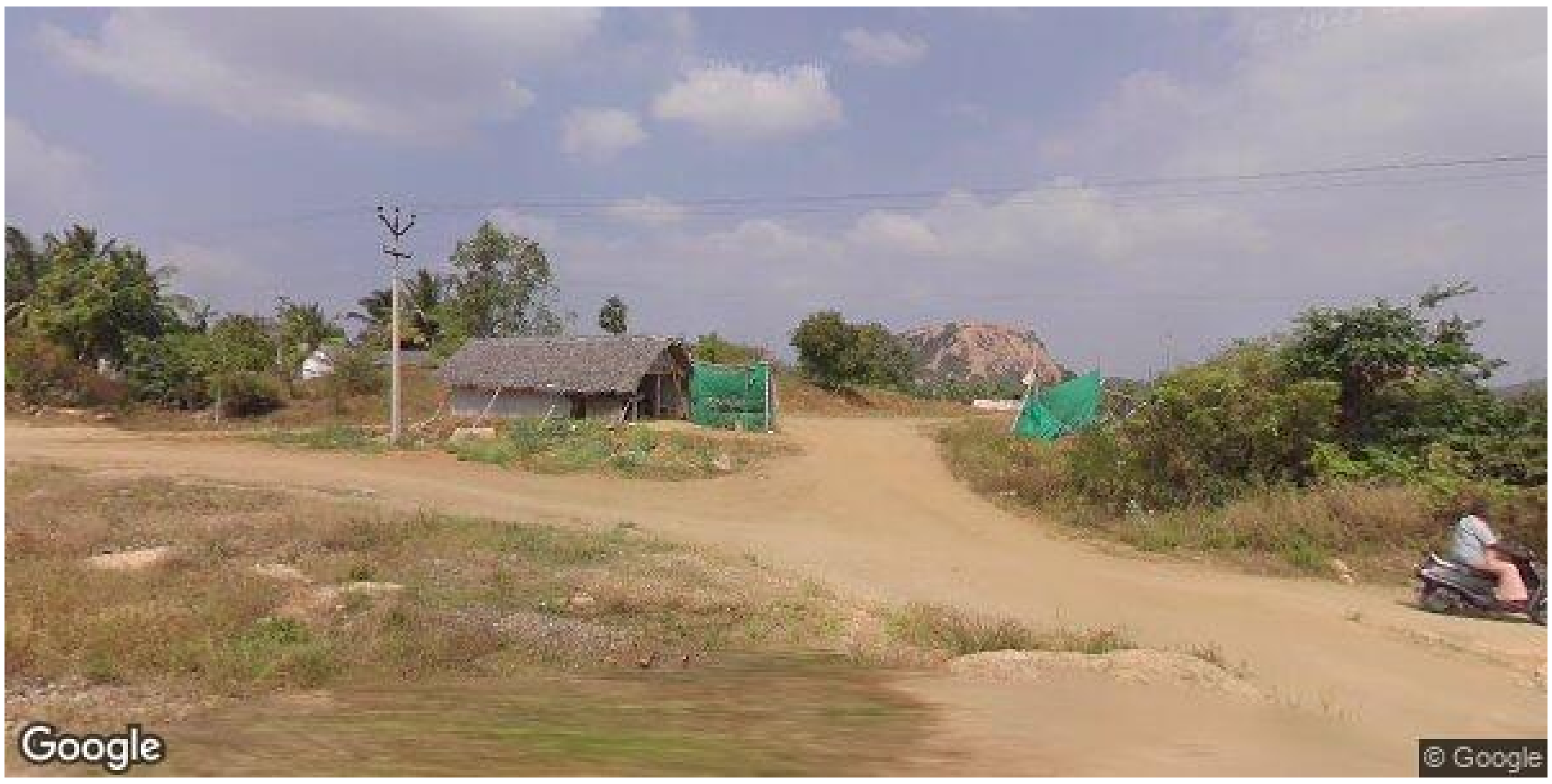
Coordinates: 11.45631, 77.81668

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49409, 77.88004



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88015

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88442

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88454

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49448, 77.88490

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.56083, 77.89921

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57048, 77.90224

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58304, 77.90656

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63840, 77.94820

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67870, 77.96846

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68905, 77.97736

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68913, 77.97734

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68925, 77.97741

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68927, 77.97747

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70223, 77.97860

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71394, 77.99572

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73661, 78.03864

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73661, 78.03886

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73772, 78.04016

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.73763, 78.04027

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.73751, 78.04016

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.73756, 78.04012

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73643, 78.03869